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A Q-learning-based Improved Adaptive Large Neighborhood Search Algorithm for Solving Agile Earth Observation Satellite Scheduling Problem

1st Fangzhou Chen
School of Automation and Electronic Information
Xiangtan University
Xiangtan , China
cfzxxf@163.com

2nd Jian Wu*
Hangzhou Institute of Technology
Xidian University
Hangzhou , China
jianwwwu@126.com

3rd Jiang Zhu*
School of Automation and Electronic Information
Xiangtan University
Xiangtan , China
zhu_jiang@xtu.ed.cn

4th Hong Ma
Key Laboratory of Intelligent Space TTC&O (Space Engineering University)
Ministry of Education
Beijing , China
mahong_0108@163.com

5th Lining Xing
Research Institute of Intelligent Control and Manufacturing System
Jiangsu University of Technology
Changzhou , China
xln@jstu.edu.cn

6th Yihui Yang
International College
Chongqing University of Posts and Telecommunications
Chongqing , China
1493368843@qq.com

***Abstract*—The scheduling problem of agile earth observation satellite (AEOS) problem presents complex constraints and high real-time requirements, inherently falling under the category of NP-hard problem. Traditional algorithms generally suffer from low efficiency. To address this, this paper propose a Q-learning-based adaptive large neighborhood search (Q-ALNS) method. This strategy leverages the destroy-and-repair process of the Adaptive Large Neighborhood Search (ALNS) algorithm and incorporates a reinforcement learning framework to dynamically select appropriate operators based on the current solution state and historical experience, thereby optimizing scheduling solutions and enhancing solution diversity as well as global search capability. Extensive experiments conducted under both global and regional distribution scenarios demonstrate that Q-ALNS significantly outperforms two state-of-the-art ALNS algorithms and the genetic algorithm in solving AEOS scheduling problem, achieving notable efficiency improvements.**

***Index Terms*—agile satellite, resource scheduling, adaptive large-domain algorithm, reinforcement learning**

I. Introduction

Over the past two decades, Earth Observation Satellite (EOS) has played an increasingly critical role as a key carrier of space information technologies in fields such as climate monitoring, resource exploration, and disaster early-warning. With the diversification of observation demands and the expansion of task scales, EOS faces challenges such as increasing scheduling complexity, rising real-time requirements, and growing difficulty in coordinating resource conflicts.

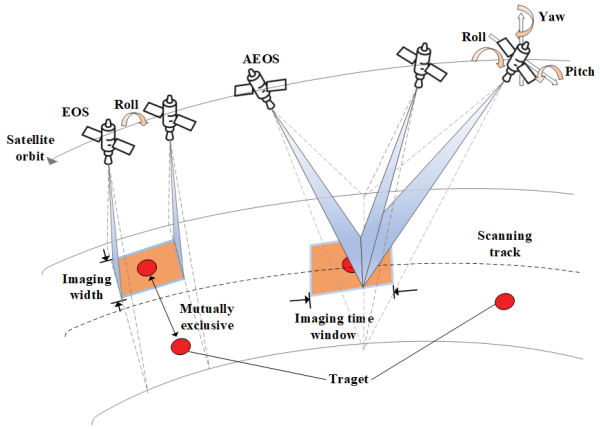


Fig. 1. The difference between AEOS and EOS

EOS task scheduling technology aims to maximize observation efficiency under limited resources and multiple constraints. Fig. 1 illustrates that traditional EOS relies solely on roll angle maneuvering, requiring imaging only when passing directly over the target, with a short and fixed visible time window (VTW) and slow task switching. In contrast, agile earth observation satellite (AEOS) exhibits maneuverability in pitch, yaw, and roll axes, allowing imaging before or after passing over the target through attitude adjustments, significantly extending the VTW and greatly improving observation efficiency and flexibility. However, the agility of AEOS significantly increases the computational complexity of the scheduling problem. Existing literature has confirmed that the AEOS scheduling problem is an NP-hard problem [1], making problem solving extremely difficult. Currently,

*Corresponding Author
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traditional algorithms for solving AEOS scheduling problem suffer from sharp performance degradation when dealing with large-scale observation tasks, highlighting the need for more advanced and efficient algorithmic solutions [2].

Current research has proposed various metaheuristic algorithms for AEOS scheduling problems, including improved quantum genetic algorithms [3], neural network-guided genetic algorithms [4] and enhanced genetic algorithms with path reconstruction operators [5]. While these methods each have their merits, they commonly suffer from issues such as high computational complexity, parameter sensitivity, and incomplete constraint modeling, making it difficult to balance solution efficiency and quality. Particularly when dealing with large-scale dynamic tasks, these algorithms demonstrate insufficient performance in terms of convergence speed and adaptability. To address these bottlenecks, research combining reinforcement learning (RL) with metaheuristic algorithms has gained increasing attention in recent years, due to its more intelligent, efficient, and adaptive nature. Recent studies [7], [8] have explored the integration of Q-learning with adaptive large neighborhood search (ALNS), where the weight update of ALNS operators is assisted by RL methods [7]. Another approach applies Q-learning to compute corresponding rewards based on the improvement in profit after each operation, demonstrating strong performance on large-scale instances [8].

Although these studies offer valuable insights into AEOS scheduling, they still face challenges such as coarse modeling, incomplete constraints, and a disconnection between strategy and state. Moreover, as scale increases, training time rises sharply and requires significant computational resources. To address these issues, this paper proposes a Q-learning-based adaptive large neighborhood search (Q-ALNS). The main contributions are as follows:

- This paper proposes an enhancement to the traditional ALNS algorithm. In standard ALNS, the algorithm dynamically adjusts the selection probabilities of different operators based on their historical performance during the iterative process. Our method innovatively incorporates a reinforcement learning mechanism, utilizing Q-learning to intelligently select operators: the algorithm learns to estimate the Q-value of each operation under given states, thereby formulating an optimal operator selection strategy. This approach significantly improves the algorithm's adaptive capabilities and effectively enhances local search efficiency.
- This paper investigates the AEOS scheduling problem, which is not only NP-hard but also possesses significant practical value. To thoroughly evaluate Q-ALNS, we designed several experimental scenarios with different distributions and scales. Comparative studies with other state-of-the-art algorithms demonstrate that our proposed method achieves superiority in both solution quality and computational efficiency.

II. Mathematical Models

A. Problem Description

In the AEOS scheduling problem, the scheduling involves multiple meta-tasks. The objective of the scheduling is to generate a feasible meta-task execution sequence that maximizes the overall profit from task completion while satisfying a set of constraints. In the mathematical model constructed in this paper, power and onboard storage constraints are temporarily excluded, and the following basic assumptions are made:

- Tasks must be executed without interruption or cancellation once initiated.
- Each satellite can execute only one task at any given time.
- Each task is treated as an independent meta-task, and may be executed only once.

B. Problem Parameter

$T = \{t_i | 1 \le i \le N\}$ denote the set of tasks;

N: the total number of tasks;

p_i: both the priority of task t_i and the profit obtained after its execution;

l_i the execution duration of task t_i;

$\gamma_t, \pi_t, \varphi_t$: the roll, pitch, and yaw angles at time t, respectively;

$w = \{w_{ij} | 1 \le i \le N, 1 \le j \le |w_i|\}$: denote the set of Visible Time Windows (VTWs);

$|w_i|$: the number of VTWs for task i;

w_{ij}: the jth VTW of task t_i;

st_{ij}, et_{ij}, d_{ij}: the start time, end time, and duration of w_{ij}, respectively.

C. Mathematical Model

The AEOS scheduling problem includes two decision variables. One is the binary variable x_{ij}, which can be expressed by Equation (1), the other is the task start time u_{ij}.

$$x_{ij} = \begin{cases} 1, & \text{if } t_i \text{ is successfully scheduled in } w_{ij} \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

where $\forall t_i \in T, w_{ij} \in w_i$. The objective function of the AEOS scheduling problem is formulated as:

$$\max \sum_{i=1}^{N} \sum_{j=1}^{|w_i|} x_{ij} p_i \tag{2}$$

Subject to the following constraints:

$$\forall t_i \in T, \sum_{j=1}^{|w_i|} x_{ij} \le 1 \tag{3}$$

$$\forall t_i \in T, st_{ij} \le u_{ij} \le u_{ij} + l_i \le et_{ij}, \text{if } x_{ij} = 1 \tag{4}$$

$$\rho_{w_{ij}, w_{i'j'}} = \begin{cases} 1, & t_i \text{ is the predecessor of } t_{i'} \\ 0, & \text{otherwise} \end{cases} \tag{5}$$

$$\Delta g = |\gamma^{w_{ij}} - \gamma^{w_{i'j'}}| + |\pi^{w_{ij}} - \pi^{w_{i'j'}}| + |\varphi^{w_{ij}} - \varphi^{w_{i'j'}}| \tag{6}$$

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$$\Delta(t_i, t_{i'}) = \begin{cases} 11.66, & \Delta g \le 10 \\ a_1 + \dfrac{\Delta g}{v_1}, & 10 < \Delta g \le 30 \\ a_2 + \dfrac{\Delta g}{v_2}, & 30 < \Delta g \le 60 \\ a_3 + \dfrac{\Delta g}{v_3}, & 60 < \Delta g \le 90 \\ a_4 + \dfrac{\Delta g}{v_4}, & \Delta g > 90 \end{cases} \tag{7}$$

$$\forall t_i, t_{i'} \in T,\ i \neq i^*,\ w_{ij} \in w_i,\ w_{i'j'} \in w_{i'},$$
$$u_{ij} + l_i + \Delta(t_i, t_{i'}) \le u_{i'j'} \quad \text{if } \rho_{w_{ij}, w_{i'j'}} = 1 \tag{8}$$

Equation (3) introduces a binary variable that constrains each task to be executed only once. Equation (4) requires that tasks be completed within the visible time window. Equation (7) establishes a piecewise function based on attitude angle variation. The values of angular velocity and fixed time constants for the corresponding angular change intervals (δg) are defined as follows: $v_1 = 1.5°/\text{s}, a_1 = 5°; v_2 = 2°/\text{s}, a_2 = 10°; v_3 = 2.5°/\text{s}, a_3 = 16°; v_4 = 3°/\text{s}, a_4 = 22°$. These parameters are utilized to compute the maneuvering time. Equation (8) requires that the interval between tasks not be shorter than the maneuvering time.

III. PROPOSED Q-ALNS ALGORITHM

A. Overall Framework

ALNS has been widely used for AEOS scheduling due to its flexible destroy-repair framework [9]. However, traditional ALNS suffers from limitations in weight adaptation, sensitivity to initial settings, and local optima convergence. Our proposed Q-ALNS overcomes these issues by integrating Q-learning to dynamically select optimal operators based on real-time Q-value evaluation, significantly enhancing search intelligence and global exploration capabilities.

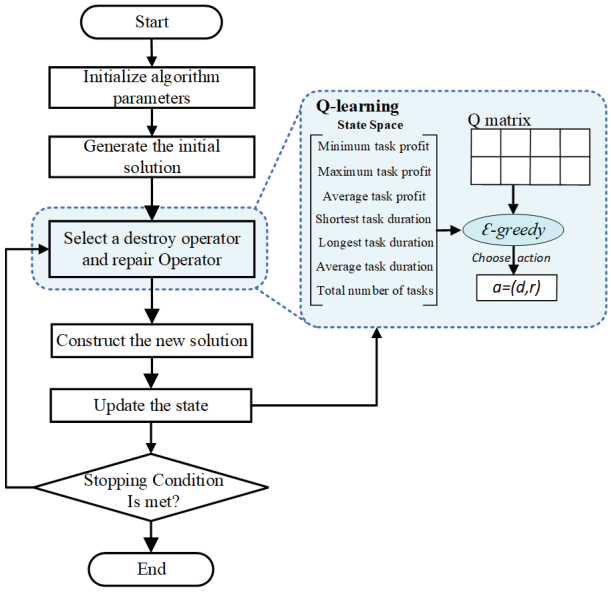


Fig. 2. Framework of Q-ALNS

As illustrated in Fig. 2, the algorithm begins with an initialization phase, generating the initial solution and operator weights. In each iteration, the Q-learning module employs an ϵ-greedy strategy to dynamically select an appropriate set of operators from the state space for destroy and repair operations. Subsequently, the Q-values are updated based on the change in solution quality after each operation, continuously refining the operator selection strategy. Finally, the output module records the best solution, improving the algorithm's global search capability while maintaining solution diversity throughout the optimization process.

B. Neighborhood Operators

This paper designs a set of specialized operators that perform local optimization on scheduling solutions through a destroy-repair mechanism. The four operators are defined as follows:

- Random Destroy Operator: Randomly removes a subset of scheduled tasks to introduce perturbations and facilitate solution space exploration.
- Low-Profit Destroy Operator: Eliminates inefficient segments of the schedule to enhance solution reconstruction quality.
- Random Repair Operator: Attempts to insert candidate tasks randomly into available scheduling gaps.
- High-Profit Repair Operator: Prioritizes high-profit tasks in a greedy manner to maximize overall returns.

In the destroy phase, the algorithm adaptively selects disturbance strategies of varying intensities based on the current solution structure and search state, leveraging neighborhood search operators. During the repair phase, a binary search-based fast insertion strategy is employed. By sorting time windows and accounting for task transition times, the algorithm efficiently locates optimal insertion positions, minimizing scheduling conflicts and idle gaps while improving resource utilization.

C. Q-Learning-Based Operator Selection

Q-learning is a model-free reinforcement learning algorithm belonging to the class of value-based temporal difference (TD) learning methods. Its fundamental principle involves constructing a Q-value function that enables the agent to iteratively update the Q-table through environmental interactions, ultimately deriving an optimal policy.

In our algorithmic framework, a pre-trained Q-table generates high-quality initial solutions by providing optimized task sequences for local search. The agent operates within a seven-dimensional state space incorporating features such as the Remaining Time Window Utilization Rate (RTUR), employing an ϵ-greedy strategy to select optimal destruction-repair operator combinations. Each action depends exclusively on the current system state, strictly adhering to Markov Decision Process (MDP) requirements.

The MDP process of Q-learning optimizes task scheduling through interactions between the agent and the environment. The agent selects appropriate actions based on the current

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state and adjusts its decisions according to the rewards received from the environment. Finally, the agent learns the Q-values for each state-action pair through the Q-table, gradually optimizing the task scheduling strategy with the goal of maximizing overall scheduling profit. The detailed process is as follows:

a) State: The state space $\mathcal{S}$ consists of a sequence of states s_t, which can be represented as:

$$s = \{s_0, s_1, ..., s_t, ...\} \tag{9}$$

Each state s_t encodes the system's current features at decision step t. s_t composed of the following feature dimensions:

$$s_t = \{RTUR_t, TPD_t, RAP_t, TWPD_t, RLB_t, TUI_t, TWR_t\} \tag{10}$$

Among them: $RTUR_t$ reflects the adequacy of the currently available observation time windows, TPD_t reflects the dispersion of the remaining task profit distribution, RAP_t directly indicates the value level of tasks to be scheduled by calculating the average profit of the remaining task set. $TWPD_t$ comprehensively considers both task value and execution duration.RLB_t shows the current utilization status of satellite resources.TUI_t dynamically reflects the pressure of scheduling progress.TWR_t reveals the matching relationship between remaining tasks and available observation windows.

$$RTUR_t = \frac{\sum_{j\in|w_i|} d_{ij}x_{ij}}{\sum_{j\in|w_i|} d_{ij}} \tag{11}$$

$$TPD_t = \frac{\sqrt{\frac{1}{N}\Sigma_{i=0}^{N}(p_i x_{ij} - \mu_p)^2}}{\mu_p} \tag{12}$$

$$RAP_t = \frac{1}{N}\sum_{i=0}^{N} p_i \tag{13}$$

$$TWPD_t = \frac{1}{N}\sum_{i=0}^{N} \frac{p_i}{l_i} \tag{14}$$

$$RLB_t = \frac{\sum_{j=0}^{|w_i|} x_{ij}l_i - t_0}{\sum_{i=0}^{N}\sum_{j=0}^{|w_i|} l_i} \tag{15}$$

$$TUI_t = \frac{\sum_{j=0}^{|w_i|} x_{ij}l_i}{\sum_{i=0}^{N}\sum_{j=0}^{|w_i|} l_i} \tag{16}$$

$$TWR_t = \frac{N}{\sum_{i=0}^{N}|w_i|} \tag{17}$$

Here, μ_p denotes the global average profit of all tasks, and t_0 represents the scheduling start time.

b) Action: The action space A consists of all possible combinations of destruction and repair operators, formally defined as: $A = D \times R$, where $D = (d_1, d_2, \ldots, d_m)$ denotes the set of destruction operators and $R = (r_1, r_2, \ldots, r_m)$ represents the set of repair operators. Action selection follows an ϵ-greedy strategy:

$$a_t = \begin{cases} ramdom(A), & \text{with probability } \epsilon \\ argmax_a Q(s,a), & \text{otherwise} \end{cases} \tag{18}$$

Where $Q(s,a)$ denotes the expected cumulative reward when executing action a in state s, as parameterized by the Q-table.

c) Reward: The reward function is formulated as:

$$R = \alpha \cdot \frac{\Delta p}{p_{max}} + \beta \cdot \frac{\sum_{i=0}^{N} x_{ij}}{N} - \gamma \cdot \frac{C_v}{C} \tag{19}$$

where Δp represents the immediate profit gain, $\frac{\sum_{i=0}^{N} x_{ij}}{N}$ calculates the task acceptance rate, and $\frac{C_v}{C}$ quantifies time constraint violations (with C_v denoting the total violation magnitude). The system dynamically evaluates solution quality changes, with rewards clipped via $clip(R, -R_{\max}, R_{\max})$ to ensure bounded values. Here, the weights α, β, γ balance trade-offs between profit maximization, task acceptance, and constraint satisfaction, respectively. A higher α prioritizes short-term profit, β emphasizes task completion, and γ penalizes violations more severely, guiding the agent toward feasible, optimal policies.

d) Value Function: Q-value updates follow the Bellman equation with ϵ-greedy exploration:

$$Q(s,a_t) = (1-\alpha)Q(s,a_t) + \alpha(R + \gamma max Q(s', a'_t)) \tag{20}$$

In this formulation, s denotes the current scheduling state, where each action $a_t = (d, r)$ combines destruction operator d and repair operator r. The Q-function $Q(s, a_t)$ estimates the long-term value of this operator pair, with the learning rate $\alpha \in (0, 1]$ controlling the weight of new experience updates. The immediate reward R and discount factor $\gamma \in [0, 1]$ respectively quantify short-term gains and future reward importance. Through iterative environment interactions, the algorithm progressively elevates Q-values for high-performing operator combinations, developing an adaptive state-dependent operator selection policy that converges toward optimal scheduling decisions.

During the training process, the Q-table maintains a matrix representation of state-action pairs, initialized either to zero (indicating no prior knowledge) or to heuristic values through pretraining. The pretraining phase incorporates domain knowledge to improve initial solution quality and accelerate convergence. During iterative exploration, the agent dynamically updates Q-values according to received rewards and observed state transitions. This hybrid pretraining-online learning approach enhances training efficiency, where pretraining provides superior initial solutions while online interactions progressively optimize the policy.

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Algorithm 1: Q-learning Training Method

Input: Tasks set T, destroy operators set D, repair operators set R, objective function f, exploration rate ϵ, learning rate α, discount rate γ, a stopping condition

Output: Trained Q-table Q

Initialize algorithm parameters and Q-table Q
$x \leftarrow$ Generate an initial solution
$s \leftarrow$ Get the initial state
for $i \leftarrow 1$ **to** M **do**
  if $random(0,1) < \epsilon$ **then**
    $(d, r) \leftarrow$ Randomly select a destroy operator and a repair operator
  else
    $(d, r) \leftarrow$ Select a destroy operator and a repair operator with the highest Q-value in the current state $\max Q(s, d, r)$
  end
  $x' \leftarrow$ Construct the new solution $r(d(x))$
  if $f(x') \geq f(x)$ **then**
    $x \leftarrow x'$
  end
  $s' \leftarrow$ Get the new state
  $Q(s, d, r) = (1-\alpha)Q(s, d, r) + \alpha(R + \gamma \max Q(s', d', r'))$
  $s \leftarrow s'$
end
return Q

IV. Experiments and Results

All experiments were conducted in an environment configured with an AMD Ryzen 7 8745H processor, 16GB RAM, NVIDIA GeForce RTX 4060 (8GB) GPU, and Python 3.9.21.

A. Experiment Settings

The parameters of the Q-ALNS algorithm are set as follows: learning rate $\alpha = 0.2$, discount factor $\gamma = 0.9$, initial exploration rate $\epsilon = 0.7$, and a decay factor of 0.99. A temperature control mechanism from simulated annealing is incorporated to gradually optimize the scheduling solution through multiple iterations. Following the methodology proposed by Wu et al. [10], we design 20 experimental scenarios with task scales ranging from 100 to 1000 in increments of 100. These scenarios are categorized into three groups based on scale: small, medium, and large. For each scale category, two distribution types are considered: global distribution and regional distribution. For comparison, we select five state-of-the-art algorithms: the tabu search-based Adaptive Large Neighborhood Search algorithm (ALNS/TPF) [9], the improved Adaptive Large Neighborhood Search algorithm (ALNS/I) [12], Duration-Sort (DS), Profit-Sort (PS), and the Genetic Algorithm (GA).

This paper designed a profit function as follows:

$$R_{region} = \frac{\sum_{i \in T_v} p_i \cdot x_{ij}}{\sum_{i \in T} p_i} \tag{21}$$

In the above formula, T_v denotes the set of tasks with available visible time windows, and p_i represents the priority of task i. The binary decision variable $x_{ij} \in 0, 1$ indicates whether task i has been successfully scheduled. For complete mathematical formulation, refer to Section II: Mathematical Model.

B. Result

Fig. 3 and Fig. 4 illustrate algorithm performance variations across different scales and distribution scenarios, using profit rate (%) as the evaluation metric (y-axis) versus task volume (x-axis). Q-ALNS demonstrates superior performance in regional distribution scenarios while maintaining competitiveness in global scenarios. This advantage stems from its dynamic Q-table updating mechanism, which intelligently selects high-value destroy/repair operations during iterations, particularly excelling in large-scale, diverse distribution scenarios.

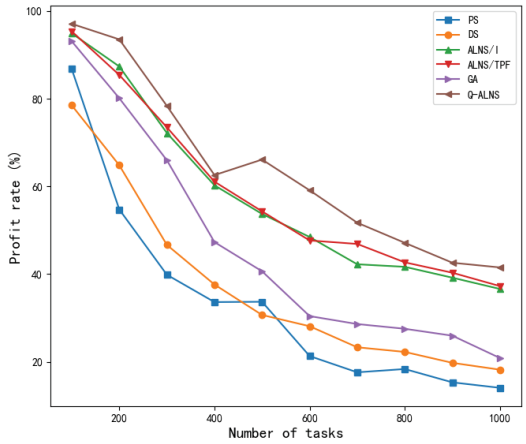


Fig. 3. Profit Variation in Regional Scenarios

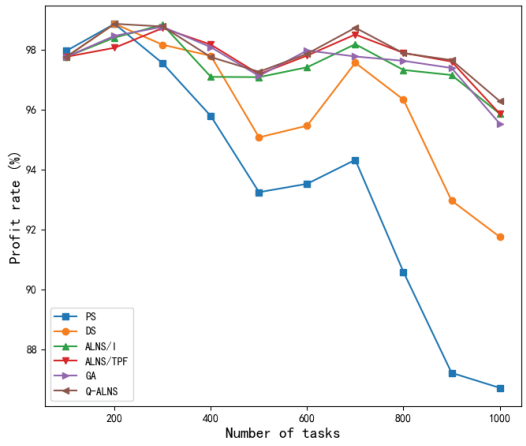


Fig. 4. Profit Variation in Global Scenarios

We evaluate four scenarios (700-1000 tasks) to assess algorithmic stability under increasing computational demands. Q-ALNS demonstrates superior stability with both highest mean

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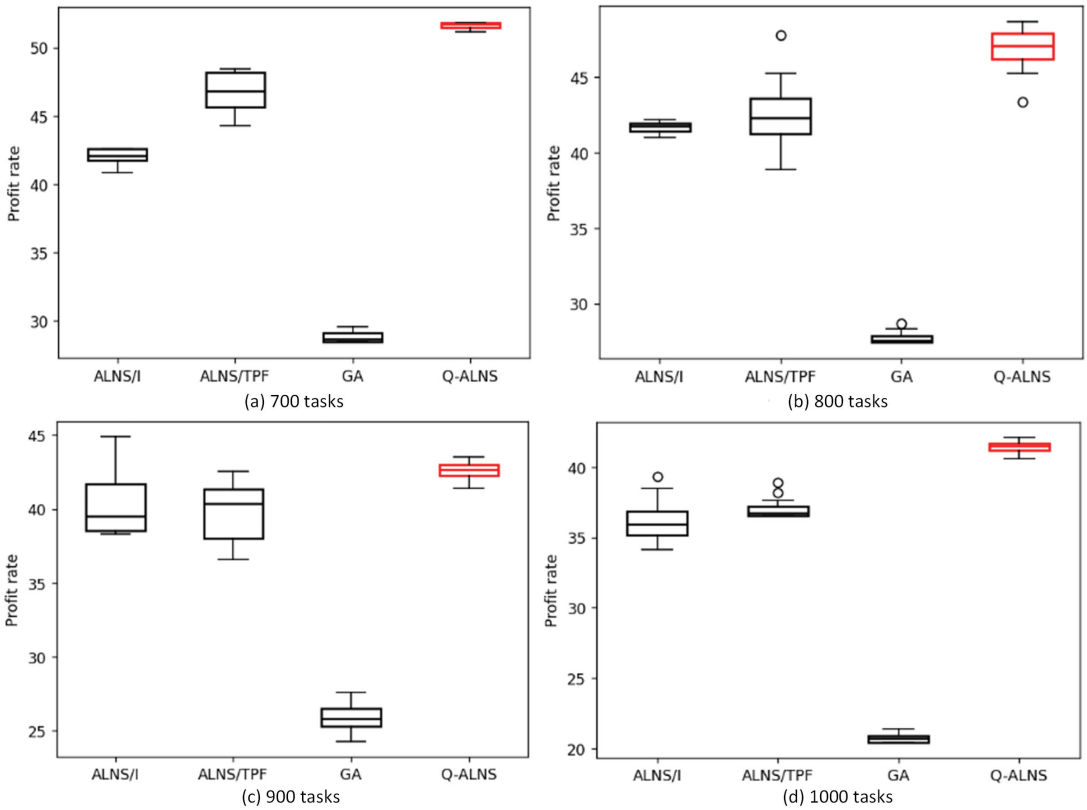


Fig. 5. Stability in different scenarios

performance and minimal variability (compact boxplots, few outliers). In contrast, ALNS/I and ALNS/TPF show increasing fluctuations with scale, while GA performs consistently worse in both metrics.

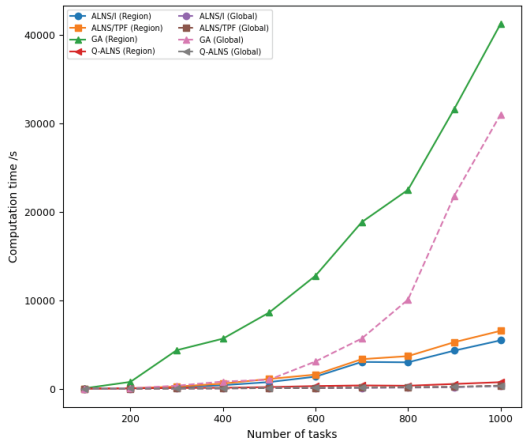


Fig. 6. CPU time (s) for each scenario

Fig. 6 compares computational efficiency across algorithms. All methods show increasing CPU time consumption with scale, with GA exhibiting the most rapid growth (lowest efficiency), followed by ALNS/TPF. Q-ALNS maintains the most gradual runtime increase while achieving the shortest overall computation time, validating its exceptional efficiency and scalability.

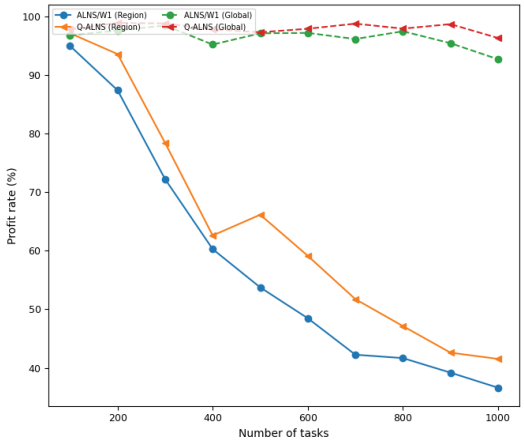


Fig. 7. Impact of Q-learning on task profit rate

To verify RL's contribution in Q-ALNS, we compare against a non-RL variant (Q-ALNS/W1). Fig. 7 confirms the Q-learning-enhanced version achieves significantly higher profits, demonstrating RL's substantial performance improvement. These results highlight Q-learning's effectiveness in identi-

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fying optimal operator combinations within complex search spaces.

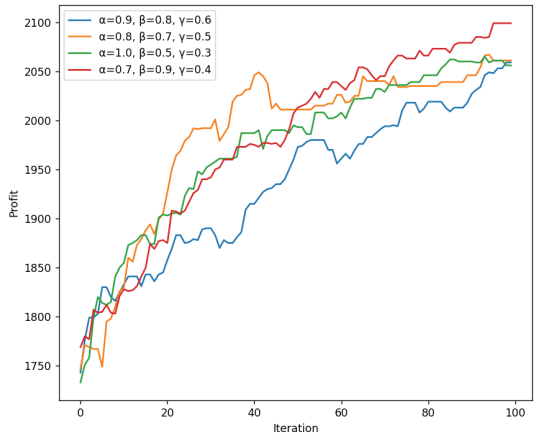


Fig. 8. Parameter sensitivity at 1000 tasks scale

Fig. 8 demonstrates Q-ALNS's profit growth across different parameter combinations. Through comprehensive parameter analysis, we examined various (α, β, γ) configurations, where α governs the exploration-exploitation trade-off, β modulates the learning rate, and γ controls reward discounting. Although certain parameter sets (e.g., $\alpha = 0.5$, $\beta = 0.8$) facilitated quicker initial convergence, the ultimate profit variations remained minimal (<2% difference). The combination ($\alpha = 0.7$, $\beta = 0.9$, $\gamma = 0.4$) yielded optimal performance, achieving 85-87% of the theoretical maximum profit. These results underscore Q-ALNS's robust adaptability, with all tested configurations demonstrating consistent profit improvement after adequate iterations. Notably, our parameter sensitivity analysis revealed that β values between 0.8-0.9 significantly improved algorithm stability during later optimization stages.

V. Conclusion

The Q-ALNS algorithm demonstrates robust performance and stability in AEOS scheduling applications, showing substantial practical potential for real-world deployment. Future research will focus on extending this framework through multi-agent Q-learning implementations for multi-satellite cooperative scheduling, incorporating distributed coordination mechanisms and adaptive reward-sharing protocols to handle complex operational scenarios, while exploring hybrid architectures that combine reinforcement learning with complementary metaheuristics like genetic algorithms and simulated annealing to enhance both solution quality and computational efficiency in dynamic scheduling environments with strict real-time constraints.

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